Problem 4 | Problems T5_MechanicalProperties.pdf #4 | Tensile test · rectangular brass sample

Tensile Test of a Rectangular Brass Sample: Modulus, Strengths, Resilience and Working Stress

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1. A tensile test was performed on a brass sample with a rectangular section and dimensions of 3.2 mm x 19.1 mm. Using the data in the table below, answer the following questions:

- *a. Represent the data in terms of engineering stress versus strain.*
- *b. Calculate the modulus of elasticity (Young's modulus).*
- *c. Determine the yield strength.*
- *d. Determine the tensile strength.*
- *e. Calculate the modulus of resilience.*
- *f. Calculate the working stress of this material.*

Point	Load (N)	Length (mm)	σn (MPa)	εn
0	0	63.50		
1	1 380	63.53		
2	2 780	63.58		
3	5 630	63.63		
4	7 430	63.70		
5	8 140	63.75		
6	9 870	64.14		
7	12 850	65.41		
8	14 100	66.68		
9	14 300	67.95		
10	13 830	69.22		
11	12 500	70.49		

(The sheet prints the table in two rows, load and length, with two more rows, σn and εn, left blank to be completed. Point 0 is the unloaded sample.)

Official Solution: **b.** $E \approx 44$ **GPa;** **c.** $\sigma_y = 150$ **MPa;** **d.** $\sigma_{UTS} = 234$ **MPa;** **e.** $U_r = 0.28$ **MPa;** **f.** $\sigma_w = 75$ **MPa**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** The section is a rectangle 3.2×19.1 mm, and up to the necking the stress is uniform: $\sigma_n = F/A_0$ and $\varepsilon_n = (l - l_0)/l_0$ [Session 11 Slide 9].
2. **Initial length.** The first value of the length row, 63.50 mm (load 0), is l_0.
3. **Linear elastic start**, Hooke's law, with the slope fitted by least squares through the origin [Session 11 Slide 13].
4. **0.2 % offset yield strength** [Session 11 Slide 28].
5. **Working stress** $\sigma_w = \sigma_y/N$ with the usual $N = 2$, which the statement does not give [Session 11 Slide 29].
6. **Resolution of the length.** The length is given to 0.01 mm, so the first elongations (0.03 and 0.08 mm) carry an error of 10 to 30 % of their value. This matters for the modulus, and it is analysed in item b.

Data

Quantity	Symbol	Value	In SI
Section	$b \times h$	19.1×3.2 mm	$19.1 \times 10^{-3} \times 3.2 \times 10^{-3}$ m
Initial length	l_0	63.50 mm	63.50×10^{-3} m
Load, length	F, l	table	N, mm
Safety factor	N	2	

Unknowns: the curve (a), E (b), σ_y (c), σ_{UTS} (d), U_r (e), σ_w (f).

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the sample), uniaxial stress, no change of basis.

Strategy. The table is turned into σ_n and ε_n. Then E is the least-squares slope through the origin of the straight part, σ_y comes from the offset construction on the segment that crosses the offset line, σ_{UTS} is the maximum load over A_0, $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35] and $\sigma_w = \sigma_y/N$. Since the first points of this table are not perfectly straight, the result is also shown for different choices of points, to see how much it depends on the choice.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = b\,h = 19.1 \text{ mm} \times 3.2 \text{ mm} = 61.12 \text{ mm}^2$$

For point 1: $F_1 = 1\,380$ N and $l_1 = 63.53$ mm, so

$$\sigma_{n,1} = \frac{1\,380\ \text{N}}{61.12\ \text{mm}^2} = 22.58\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{63.53 - 63.50}{63.50} = 0.00047$$

a) Engineering stress and strain, and the curve

All the points:

Point	F (N)	l (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
0	0	63.50	0	0.00000	—
1	1 380	63.53	22.58	0.00047	47.8
2	2 780	63.58	45.48	0.00126	36.1
3	5 630	63.63	92.11	0.00205	45.0
4	7 430	63.70	121.56	0.00315	38.6
5	8 140	63.75	133.18	0.00394	33.8
6	9 870	64.14	161.49	0.01008	16.0
7	12 850	65.41	210.24	0.03008	7.0
8	14 100	66.68	230.69	0.05008	4.6
9	14 300	67.95	233.97	0.07008	3.3
10	13 830	69.22	226.28	0.09008	2.5
11	12 500	70.49	204.52	0.11008	1.9

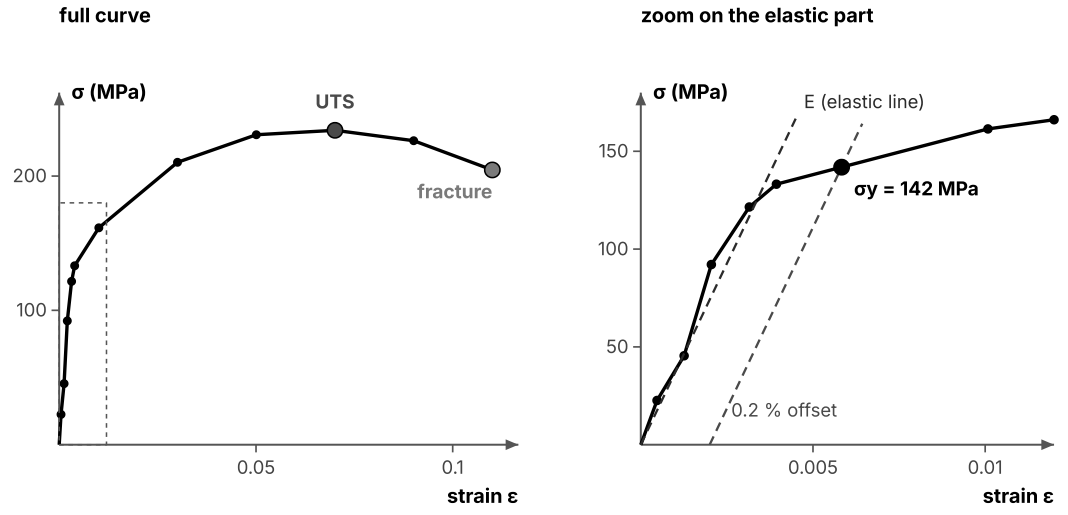


Figure 4. Problem 4: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Modulus of elasticity

The secant modulus of the last column is 34 to 48 GPa for points 1 to 5 and falls to 16 GPa at point 6: the straight part ends after point 5. The scatter between 34 and 48 GPa comes from the resolution of the length: point 1 is an elongation of 0.03 mm, and an error of 0.01 mm is 33 % of it. A fit through the origin on points 1 to 5 weights the points by their strain, so the points with larger elongation, and smaller relative error, count more:

Point	ε_i	σ_i (MPa)	$\sigma_i\varepsilon_i$ (MPa)	ε_i^2
1	0.00047	22.58	0.0107	2.232×10^{-7}
2	0.00126	45.48	0.0573	1.587×10^{-6}
3	0.00205	92.11	0.1886	4.191×10^{-6}
4	0.00315	121.56	0.3829	9.920×10^{-6}
5	0.00394	133.18	0.5243	1.550×10^{-5}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{1.164}{3.1422 \times 10^{-5}} = 37037 \text{ MPa} = \mathbf{37.0 \text{ GPa}}$$

c) Yield strength (0.2 % offset)

With the offset line $E(\varepsilon - 0.002)$ the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ changes between points 5 and 6:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
4	0.00315	121.6	42.6	79.0
5	0.00394	133.2	71.7	61.4
6	0.01008	161.5	299.2	-137.7
7	0.03008	210.2	1040.0	-829.7

$$m = \frac{161.49 - 133.18}{0.01008 - 0.00394} = 4608.6 \text{ MPa} \qquad \varepsilon^* = \frac{133.18 - 4608.6 \times 0.00394 + 0.002 \times 37037}{37037 - 4608.6} = \frac{189.110}{32428.3} = 0.0058316$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 37037 \times (0.0058316 - 0.002) = \mathbf{141.9 \text{ MPa}}$$

d) Tensile strength

The maximum load is $F_{max} = 14\,300$ N (point 9):

$$\sigma_{UTS} = \frac{F_{max}}{A_0} = \frac{14\,300\ \text{N}}{61.12\ \text{mm}^2} = \mathbf{234.0\ MPa}$$

e) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(141.9\ \text{MPa})^2}{2 \times 37037\ \text{MPa}} = \mathbf{0.272\ MPa}$$

f) Working stress

$$\sigma_w = \frac{\sigma_y}{N} = \frac{141.9}{2} = \mathbf{71.0\ MPa}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** N/mm² is MPa, strains have no units, E and U_r are in MPa.
- **Order.** $\sigma_y = 141.9$ MPa $< \sigma_{UTS} = 234.0$ MPa; after the maximum the load falls (point 11 has 12 500 N), the sample necks.
- **Sensitivity to the points chosen for E.** The modulus, and through it σ_y and U_r, depend on how many of the first points are taken:

Points used for E	E (GPa)	σ_y (MPa)	U_r (MPa)	σ_w (MPa)
1 to 3	42.7	139.3	0.227	69.6
1 to 4	40.2	140.4	0.245	70.2
1 to 5	37.0	141.9	0.272	71.0

Between the three choices the yield strength changes by 1.9 % and U_r by 20 % (because it contains $1/E$): the strengths hardly depend on the choice made for E, but the resilience does.

Interpretation

The brass has a tensile strength of 234.0 MPa and a yield strength of about 141.9 MPa, and between the two the strain hardening adds 92 MPa. The large strain at fracture (11 % at the last point, after the maximum load) shows that it is ductile.

THE MODULUS IS MUCH LOWER THAN THE ONE OF BRASS IN THE SLIDE

The table of Session 11 gives $E = 97$ GPa for brass [Session 11 Slide 17]. The data of this problem give 37 to 43 GPa, less than half. The key also gives 44 GPa, so the sheet itself uses these data. The reason is probably that the elongation of the table includes the slack and the compliance of the testing machine in the first points, which is why the elastic part looks softer than the real material. The data are used as they are.

Check against the official solution

Item	Official	Computed	Comment
b. E	≈ 44 GPa	37.0 GPa	differs 16 %: see the sensitivity table; with points 1 to 3 the fit gives 42.7 GPa
c. σ_y	150 MPa	141.9 MPa	differs 5 %
d. σ_{UTS}	234 MPa	234.0 MPa	agrees
e. U_r	0.28 MPa	0.272 MPa	agrees (3 %)
f. σ_w	75 MPa	71.0 MPa	consistent with σ_y

DISCREPANCY WITH THE OFFICIAL SOLUTION: MODULUS AND YIELD STRENGTH

The key gives $E \approx 44$ GPa and $\sigma_y = 150$ MPa. The 0.2 % offset on the data of the table gives σ_y between 139 MPa ($E = 44$ GPa) and 142 MPa ($E = 37.0$ GPa), never 150 MPa. A yield strength of 150 MPa corresponds to the stress where the curve bends clearly (points 5 and 6, 133 to 161 MPa), that is, to a reading by eye of the end of the straight part. In addition, with the key values, $U_r = 150^2/(2 \times 44\,000) = 0.256$ MPa, not 0.28 MPa; the computed resilience, 0.272 MPa, is close to the printed one only by coincidence. The computed values are kept.